

$$LC \frac{d^2 y}{dt^2} + RC \frac{dy}{dt} + y(t) = x(t)$$

Aplicando la transformada de Laplace con las Condiciones

$$LC [s^2 Y(s) - sy(0^+) - y'(0^+)] + RC [sy(s) - y(0^+)] + Y(s) = X(s)$$

Agrupando $Y(s)$

$$(LCs^2 + RCs + 1) Y(s) = X(s) + LCsy(0^+) + LCy'(0^+) + RCy(0^+)$$

Factorizando

$$Y(s) = \frac{X(s)}{LCs^2 + RCs + 1} + \frac{LCs + RC}{LCs^2 + RCs + 1} y(0^+) + \frac{LC}{LCs^2 + RCs + 1} y'(0^+)$$

relacionando
 $y(0^+)$

Sebenos que

$$ix(t) = c y(t)$$

$$y'(0^+) = \frac{ix(0^+)}{c}$$

Sustituimos:

$$Y(s) = \frac{X(s)}{LCs^2 + RCs + 1} + \frac{LCs + RC}{LCs^2 + RCs + 1} \cdot y(0^+) + \frac{L}{LCs^2 + RCs + 1} ix(0^+)$$

y llegamos a

$Y_{ext}(s)$

$Y_{int}(s)$

$$Y(s) = Y_{ext}(s) + Y_{int}(s)$$